

A Test of Mean Resolution Time

AP Statistics · Unit 4 · Topic 4.5 · TEACHER KEY

CED TETHER

- **Skill 3.E** — Calculate a test statistic and find a p-value, provided conditions for inference are met.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **Skill 4.E** — Justify a claim using a decision based on significance tests.
- **EU VAR-7** — The t-distribution may be used to model variation.
- **LO VAR-7.E** — Calculate an appropriate test statistic for a population mean, including the mean difference between values in matched pairs. [Skill 3.E]
- **EK VAR-7.E.1** — For a single quantitative variable when random sampling with replacement from a population that can be modeled with a normal distribution with mean μ and standard deviation σ , the sampling distribution of $t = (\bar{x} - \mu) / (s/\sqrt{n})$ has a t-distribution with $n - 1$ degrees of freedom.
- **EU DAT-3** — Significance testing allows us to make decisions about hypotheses within a particular context.
- **LO DAT-3.E** — Interpret the p-value of a significance test for a population mean, including the mean difference between values in matched pairs. [Skill 4.B]
- **EK DAT-3.E.1** — An interpretation of the p-value of a significance test for a population mean should recognize that the p-value is computed by assuming that the null hypothesis is true, i.e., by assuming that the true population mean is equal to the particular value stated in the null hypothesis.
- **LO DAT-3.F** — Justify a claim about the population based on the results of a significance test for a population mean. [Skill 4.E]
- **EK DAT-3.F.1** — A formal decision explicitly compares the p-value to the significance α . If the p-value $\leq \alpha$, then reject the null hypothesis, $H_0: \mu = \mu_0$. If the p-value $> \alpha$, then fail to reject the null hypothesis.
- **EK DAT-3.F.2** — The results of a significance test for a population mean can serve as the statistical reasoning to support the answer to a research question about the population that was sampled.

Essential question: What conclusion is warranted when a one-sample t-test does not cross its evidence threshold?

DOK-3 skill: 4.E · **FRQ pattern:** mean-test-decision-and-conclusion

DOK rationale: Part (c) coordinates the lower-tail p-value, preset threshold, asymmetric decision logic, and SRS population scope to adjudicate two conclusions and trace the consequence of overclaiming.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which specific number or study detail supports your choice?
- What claim would go beyond the evidence, and why?

[WARN] LOOK FOR

- Choosing a speaker without a reason.
- Copying numbers without explaining what they support.

SCORING PART (c) — E / P / I

Expected elements

- **decision** (required) — Uses $0.0645 > 0.05$ to fail to reject
- **context** (required) — States insufficient evidence for a mean below 12 hours among the 1,200 tickets
- **limits** (required) — Explains why the test does not prove equality or chance as the only explanation

E	Uses the threshold correctly, gives the population conclusion, and explains both limits on what the test proves.
P	Shows substantive valid reasoning for at least one required element, but does not meet all E requirements. Credit correct reasoning even when another claim is wrong.
I	Shows no substantive valid reasoning for any required element; a name, unsupported choice, or copied number alone does not count.

[STAR] THE PROBLEM — annotated

Suppose a help desk closed 1,200 tickets in a quarter. To investigate whether their population mean resolution time was below 12 hours, an analyst took an SRS of 24 and used $\alpha = 0.05$.

Leah: “The lower-tail test gives $t = -1.5747$, $df = 23$, and $p \approx 0.0645$. I would state what evidence this does and does not provide.”

Damon: “Because $p > 0.05$, accept H_0 . The test proves the population mean was exactly 12 hours and the observed difference was only chance.”

Ticket sample

Summary	Value
Population	1200
n	24
Mean (h)	11.1
SD (h)	2.8
Shape	Symmetric; no outliers

FIRST TAKE — one sentence before you work the parts

Does this small study seem able to prove that the average was exactly 12 hours? Explain in one sentence.

Key: Ungraded commitment; accept any evidence-based first impression.

[BOOK] COMPARE EVIDENCE TO THE PRESET THRESHOLD (keep on the page)

For a one-sample mean test with unknown population standard deviation, $t = (\bar{x} - \mu_0)/(s/\sqrt{n})$ has $df = n - 1$. The p -value is the probability, assuming H_0 is true, of a test statistic at least as extreme as the observed statistic in the direction of H_a . If $p \leq \alpha$, reject H_0 ; if $p > \alpha$, fail to reject H_0 . Failing to reject says the sample does not provide convincing evidence for H_a at that significance level. It does not prove the null value, and the conclusion must concern the population represented by the random sample.

DOK 1 (a) State H_0 and H_a for the population mean resolution time.

Key: $H_0 : \mu = 12$ hours; $H_a : \mu < 12$ hours, where μ is the mean for all 1,200 tickets closed that quarter.

DOK 2 (b) Using $\bar{x} = 11.1$, $s = 2.8$, and $n = 24$, calculate t . Use the supplied $p = 0.0645$; you do not need to find it again.

Key: $t = (11.1 - 12)/(2.8/\sqrt{24}) = -1.5747$. The supplied lower-tail p -value is about 0.0645 with $df = 23$.

DOK 3 (c) In 3–4 sentences, decide whose account is justified. Compare the p -value with 0.05, state the conclusion for all 1,200 tickets, and explain why failure to reject does not prove equality or that chance was the only explanation.

Key: Leah is justified: $0.0645 > 0.05$, so fail to reject H_0 . There is not convincing evidence that the mean resolution time for all 1,200 tickets was below 12 hours. The p -value measures how unusual the data would be assuming a mean of 12; it does not prove that assumption. Chance remains a possible explanation, but the test does not prove it was the only explanation.